

Multiple Choice (10 marks)

1. The body mass index (BMI) of vegetarians differs on average from meat-eaters by how much?
 - (a) 3 units of BMI
 - (b) 0.5 units of BMI
 - (c) 1 unit of BMI
 - (d) 2 units of BMI
2. Which of the following statements about glucose metabolism is correct?
 - (a) Glucagon increases the rate of glycolysis
 - (b) Glycolysis requires NADP+
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) The end-product of glycolysis in red blood cells is pyruvate
3. Which vitamin is required for synthesis of the blood clotting proteins?
 - (a) Vitamin A
 - (b) Vitamin D
 - (c) Vitamin E
 - (d) Vitamin K
4. Observational epidemiology studies have produced useful in identifying health effects of many food contaminants, but not for assessing risk, because:
 - (a) It is never possible to know whether observed effects are caused by the contaminant
 - (b) Human exposure through food is usually much less than exposures that have been studied
 - (c) Controlled animal studies provide more reliable information
 - (d) Reliable dose-response data are difficult to acquire and often unavailable
5. Which of the following will result in an increase in the rate of tissue protein synthesis?
 - (a) Increased glucagon secretion in the fasting state.
 - (b) Increased insulin secretion in the fed state.
 - (c) Increased availability of lysine in the fed state.
 - (d) Secretion of cortisol under conditions of stress.

True / False (10 marks)

Answer True or False

6. True or False: The complete oxidation of glucose to carbon dioxide and water yields 32 molecules of ATP.
7. True or False: Low-carbohydrate diets were previously not recommended because they resulted in less weight loss efficacy.
8. True or False: The impact of 'mad cow disease' on food choice is largely due to a difference in perception of risk among consumers.
9. True or False: VLCD provides less energy than LCD and usually does not include natural foods.
10. True or False: A sedentary 43-year-old woman who weighs 63 kg needs about 1250 kcal per day for her energy needs.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between obesity and endometrial cancer, focusing on the role of hormones, particularly oestrogen, in mediating this risk.
12. Discuss the key differences between type 1 and type 2 diabetes, focusing on the role of autoimmune factors in their development and how these differences impact management and treatment.

End of Paper